Trajectory Smoothing And Avoidance Literature Review

Fifteen recent papers selected for a JSSP/RL autonomous intersection manager with low-level smooth trajectory execution. Review date: 2026-06-26.

The list balances stop avoidance, energy/delay reduction, collision avoidance, graph scheduling, RL, and replanning.

15
Core papers

83.5
Average relevance

6
High-fit papers

15
Open PDFs

P01 · 2021 · Preprint · relevance 94/100

A Priority-Aware Replanning and Resequencing Framework for Coordination of Connected and Automated Vehicles

Behdad Chalaki; Andreas A. Malikopoulos

Venue
arXiv preprint

Theme
Scheduling/resequencing + optimal control

Source
https://arxiv.org/abs/2109.05573

PDF
https://arxiv.org/pdf/2109.05573
open PDF downloaded from arXiv

Technical Summary

Event-driven replanning and priority-aware resequencing layered on decentralized CAV optimal control at a signal-free intersection.

Stop Avoidance

The lower-level optimal-control trajectory is constructed so vehicles can conserve momentum rather than stop at the intersection when a feasible exit time exists.

Collision Avoidance

Uses rear-end and lateral safety constraints around conflict points, with replanning when disturbances change feasibility.

Validation

Numerical simulation of signal-free intersection coordination under resequencing and replanning.

Limitations

Assumes connected automated vehicles, known paths, and simplified dynamics; the paper is closer to resequencing theory than to learned graph scheduling.

Use In This Project

Directly supports this project's hierarchy: a high-level sequence or schedule provides target times, and a low-level controller checks executable motion.

Venue/source note: Useful technical preprint from a well-known CAV optimal-control group; venue reputation not verified beyond arXiv.

P02 · 2021 · Journal / Preprint · relevance 92/100

Conflict-free Cooperation Method for Connected and Automated Vehicles at Unsignalized Intersections: Graph-based Modeling and Optimality Analysis

Chaoyi Chen; Qing Xu; Mengchi Cai; Jiawei Wang; Jianqiang Wang; Biao Xu; Keqiang Li

Venue
Transportation Research Part C: Emerging Technologies / arXiv version

Theme
Graph conflict modeling + passing order

Source
https://arxiv.org/abs/2107.07179

PDF
https://arxiv.org/pdf/2107.07179
open PDF downloaded from arXiv

Technical Summary

Models trajectory conflicts as a conflict directed graph and coexisting undirected graph; solves passing order through improved depth-first spanning tree and minimum clique cover methods.

Stop Avoidance

Optimizes evacuation time and average delay, using distributed control to reduce idling near the stop line.

Collision Avoidance

Conflict-free graph edges represent crossing, diverging, converging, and reachability conflicts before scheduling.

Validation

Traffic simulations over varying vehicle counts and volumes; reports efficiency and fuel-economy improvements.

Limitations

Mostly longitudinal control, ideal communication, and no lane changing in this paper's base formulation.

Use In This Project

Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

Venue/source note: Transportation Research Part C is a leading transportation automation journal; exact publication metadata should be checked before final submission.

P03 · 2021 · Preprint · relevance 82/100

Cooperation Method of Connected and Automated Vehicles at Unsignalized Intersections: Lane Changing and Arrival Scheduling

Chaoyi Chen; Mengchi Cai; Jiawei Wang; Kai Li; Qing Xu; Jianqiang Wang; Keqiang Li

Venue
arXiv preprint

Theme
Lane changing + arrival scheduling

Source
https://arxiv.org/abs/2109.14175

PDF
https://arxiv.org/pdf/2109.14175
open PDF downloaded from arXiv

Technical Summary

Two-stage framework decoupling lateral lane assignment/path planning from graph-based arrival scheduling with minimum clique cover.

Stop Avoidance

Improves traffic efficiency by choosing target lanes and arrival schedules that avoid unnecessary queuing.

Collision Avoidance

Conflict-free arrival scheduling and formation-control assumptions prevent incompatible movements.

Validation

Numerical simulations for different vehicle counts and traffic volumes.

Limitations

arXiv note reports text overlap with the authors' earlier graph-based paper; use mainly for lane-changing extension context.

Use In This Project

Useful for discussing route-template/lane-choice extensions beyond the current fixed-route JSSP formulation.

Venue/source note: arXiv preprint; venue reputation not verified.

P04 · 2023 · Journal · relevance 91/100

Real-time Cooperative Vehicle Coordination at Unsignalized Road Intersections

Jiping Luo; Tingting Zhang; Rui Hao; Donglin Li; Chunsheng Chen; Zhenyu Na; Qinyu Zhang

Venue
IEEE Transactions on Intelligent Transportation Systems

Theme
DRL cooperative trajectory optimization

Source
https://arxiv.org/abs/2205.01278

PDF
https://arxiv.org/pdf/2205.01278
open PDF downloaded from arXiv

Technical Summary

Formulates unified cooperative trajectory optimization for unsignalized intersections and solves the sequential decision problem with TD3.

Stop Avoidance

Optimizes throughput and continuous flow, reducing delay compared with conservative coordination.

Collision Avoidance

Safety and long-term stability constraints are embedded in the cooperative trajectory optimization framework.

Validation

Simulation and practical experiments; authors report millisecond-scale computation and scalability with lane count.

Limitations

Centralized control-authority handoff and DRL policy behavior may be difficult to certify in mixed traffic.

Use In This Project

Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

Venue/source note: IEEE T-ITS is a highly reputable ITS journal; DOI and journal reference verified on arXiv.

P05 · 2022 · Journal · relevance 86/100

Hybrid Reinforcement Learning-Based Eco-Driving Strategy for Connected and Automated Vehicles at Signalized Intersections

Zhengwei Bai; Peng Hao; Wei Shangguan; Baigen Cai; Matthew J. Barth

Venue
IEEE Transactions on Intelligent Transportation Systems

Theme
Eco-driving + hybrid RL

Source
https://arxiv.org/abs/2201.07833

PDF
https://arxiv.org/pdf/2201.07833
open PDF downloaded from arXiv

Technical Summary

Combines a rule-based driving manager, V2I/vision features, and RL to generate longitudinal and lateral eco-driving actions.

Stop Avoidance

Targets energy and travel-time reduction by smoothing approach behavior at signalized intersections.

Collision Avoidance

Mixed-traffic policy includes rule-based safety management around surrounding vehicles.

Validation

Unity-based mixed-traffic intersection simulation; reports 12.70 percent energy reduction and 11.75 percent travel-time saving against a model-based eco-driving baseline.

Limitations

Signalized setting and policy-level action output differ from a scheduler-to-smoother architecture.

Use In This Project

Supports the argument that stop-and-go reduction and energy-aware smoothing are important low-level objectives.

Venue/source note: IEEE T-ITS is a highly reputable ITS journal; DOI and journal reference verified on arXiv.

P06 · 2023 · Journal / Preprint · relevance 83/100

Eco-driving for Electric Connected Vehicles at Signalized Intersections: A Parameterized Reinforcement Learning Approach

Xia Jiang; Jian Zhang; Dan Li

Venue
Journal article related DOI / arXiv version

Theme
Electric CV eco-driving + parameterized RL

Source
https://arxiv.org/abs/2206.12065

PDF
https://arxiv.org/pdf/2206.12065
open PDF downloaded from arXiv

Technical Summary

Parameterized RL combines model-based car-following, lane-changing policy, and learned longitudinal/lateral decisions.

Stop Avoidance

Learns energy-saving actions around signalized intersections without interrupting surrounding HDVs.

Collision Avoidance

Safety is handled through model-based car-following and lane-changing constraints around human-driven vehicles.

Validation

SUMO evaluation from single-vehicle and flow perspectives.

Limitations

Signalized intersections and electric connected vehicles rather than fully signal-free autonomous scheduling.

Use In This Project

Provides a SUMO-compatible example of energy and stop reduction metrics for the experimental section.

Venue/source note: Related DOI exists on arXiv; exact journal reputation not verified in this pass.

P07 · 2022 · Conference / Preprint · relevance 81/100

Cooperative Control in Eco-Driving of Electric Connected and Autonomous Vehicles in an Un-Signalized Urban Intersection

Vinith Kumar Lakshmanan; Antonio Sciarretta; Ouafae El Ganaoui-Mourlan

Venue
AAC 2022 submission / arXiv version

Theme
Eco-driving optimal speed profile

Source
https://arxiv.org/abs/2206.12360

PDF
https://arxiv.org/pdf/2206.12360
open PDF downloaded from arXiv

Technical Summary

Single-level eco-driving optimization solved with Pontryagin's Minimum Principle for electric CAV speed profiles at an unsignalized intersection.

Stop Avoidance

Focuses directly on optimal speed profiles that reduce energy use and avoid inefficient stop-and-go behavior.

Collision Avoidance

Analytical conflict cases and cooperation levels encode interaction safety among CAVs.

Validation

Simulation comparison of cooperative and non-cooperative eco-driving against IDM baseline.

Limitations

Narrower single-vehicle/eco-driving emphasis; less detail on high-level scheduling.

Use In This Project

Supports the smoothing objective terms for acceleration, energy proxy, and stop avoidance.

Venue/source note: Conference/preprint status only; venue reputation not verified.

P08 · 2022 · Preprint · relevance 78/100

Safety-Aware and Data-Driven Predictive Control for Connected Automated Vehicles at a Mixed Traffic Signalized Intersection

A. M. Ishtiaque Mahbub; Viet-Anh Le; Andreas A. Malikopoulos

Venue
arXiv preprint

Theme
Data-driven predictive control + mixed traffic safety

Source
https://arxiv.org/abs/2203.05739

PDF
https://arxiv.org/pdf/2203.05739
open PDF downloaded from arXiv

Technical Summary

Finite-horizon predictive control uses recursive least squares to estimate preceding HDV behavior in real time.

Stop Avoidance

Smooths CAV response near yellow/red phases while avoiding overly conservative braking behind HDVs.

Collision Avoidance

Prioritizes rear-end collision avoidance when preceding human-driven vehicles approach signal changes.

Validation

Numerical simulation and robustness analysis.

Limitations

Signalized and rear-end-focused rather than conflict-zone scheduling across all movements.

Use In This Project

Useful for fallback and mixed-traffic safety discussion, especially when predicted trajectories deviate.

Venue/source note: arXiv preprint; venue reputation not verified.

P09 · 2023 · Preprint · relevance 84/100

Overtaking-enabled Eco-approach Control at Signalized Intersections for Connected and Automated Vehicles

Haoxuan Dong; Weichao Zhuang; Guoyuan Wu; Zhaojian Li; Guodong Yin; Ziyou Song

Venue
arXiv preprint

Theme
Eco-approach + relaxed FIFO

Source
https://arxiv.org/abs/2306.09736

PDF
https://arxiv.org/pdf/2306.09736
open PDF downloaded from arXiv

Technical Summary

Receding-horizon two-stage control: dynamic-programming lane optimization followed by PMP speed trajectory optimization.

Stop Avoidance

Relaxes FIFO queuing at signalized intersections to reduce driving cost, energy consumption, and delay.

Collision Avoidance

Handles dynamic disturbance from preceding vehicles and lane-choice constraints.

Validation

Extensive simulations report average driving-cost improvements over constant-speed and regular eco-approach baselines.

Limitations

Signalized intersection and overtaking/lane planning assumptions are outside the current fixed-route JSSP scope.

Use In This Project

Supports the critique that FCFS/FIFO can be suboptimal and that speed smoothing should be coupled to sequencing.

Venue/source note: arXiv preprint; venue reputation not verified.

P10 · 2026 · Magazine / Preprint · relevance 90/100

Real-World Evaluation of Two Cooperative Intersection Management Approaches

Marvin Klimke; Max Bastian Mertens; Benjamin Volz; Michael Buchholz

Venue
IEEE Intelligent Transportation Systems Magazine (accepted)

Theme
Real-world cooperative intersection management

Source
https://arxiv.org/abs/2403.16478

PDF
https://arxiv.org/pdf/2403.16478
open PDF downloaded from arXiv

Technical Summary

Compares multi-scenario prediction and graph-based reinforcement learning approaches in mixed-traffic simulation and real drives.

Stop Avoidance

Reports substantial reductions in crossing time and number of stops for cooperative maneuver planning.

Collision Avoidance

Evaluates criticality metrics while operating with prototype connected automated vehicles in public traffic.

Validation

Novel mixed-traffic simulation framework plus real-world prototype connected automated vehicle drives.

Limitations

Focuses on cooperative maneuver planning evaluation, not a JSSP formulation.

Use In This Project

Excellent evidence that stop count is a practical metric and that real-world mixed traffic should be considered in future validation.

Venue/source note: IEEE ITS Magazine is reputable and practice-facing; accepted-publication note and DOI are verified on arXiv.

P11 · 2025 · Preprint · relevance 92/100

A Spatial-Domain Coordinated Control Method for CAVs at Unsignalized Intersections Considering Motion Uncertainty

Tong Zhao; Nikolce Murgovski; Baigen Cai; Wei ShangGuan

Venue
arXiv preprint

Theme
Spatial-domain robust trajectory planning

Source
https://arxiv.org/abs/2412.04290

PDF
https://arxiv.org/pdf/2412.04290
open PDF downloaded from arXiv

Technical Summary

Transforms coordinated control into a spatial-domain nonlinear program with unified linear collision-avoidance constraints and real-time iteration.

Stop Avoidance

Generates smooth trajectories under state/control constraints, avoiding unnecessary braking where robust constraints permit.

Collision Avoidance

Handles crossing, following, merging, and diverging conflicts under path and speed uncertainty of HDVs.

Validation

Simulation case studies; reported RTI computation reduction by orders of magnitude with small optimality loss.

Limitations

Optimization-heavy; not a learned scheduler, and publication venue beyond arXiv not verified.

Use In This Project

Very strong for the smoother's feasibility and robust collision-avoidance constraints.

Venue/source note: Recent technical arXiv preprint; venue reputation not verified.

P12 · 2025 · Preprint · relevance 84/100

Uncertainty-Aware Safety-Critical Decision and Control for Autonomous Vehicles at Unsignalized Intersections

Ran Yu; Zhuoren Li; Lu Xiong; Wei Han; Bo Leng

Venue
arXiv preprint

Theme
Uncertainty-aware safe RL + HOCBF

Source
https://arxiv.org/abs/2505.19939

PDF
https://arxiv.org/pdf/2505.19939
open PDF downloaded from arXiv

Technical Summary

Risk-aware ensemble distributional RL estimates policy uncertainty, then a high-order control barrier function acts as a safety filter.

Stop Avoidance

Balances safety and traffic efficiency by switching between flexible RL behavior and conservative HOCBF intervention.

Collision Avoidance

HOCBF safety filter dynamically adjusts constraints using joint uncertainty in unsignalized-intersection tasks.

Validation

Simulation tests on multiple unsignalized-intersection tasks against safety and efficiency baselines.

Limitations

Single-agent urban autonomous-driving framing rather than centralized conflict-zone scheduling.

Use In This Project

Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

Venue/source note: Recent arXiv preprint; venue reputation not verified.

P13 · 2023 · Survey / Preprint · relevance 72/100

A Systematic Survey of Control Techniques and Applications in Connected and Automated Vehicles

Wei Liu; Min Hua; Zhiyun Deng; Zonglin Meng; Yanjun Huang; Chuan Hu; Shunhui Song; Letian Gao; Changsheng Liu; Bin Shuai; Amir Khajepour; Lu Xiong; Xin Xia

Venue
arXiv survey

Theme
CAV control survey

Source
https://arxiv.org/abs/2303.05665

PDF
https://arxiv.org/pdf/2303.05665
open PDF downloaded from arXiv

Technical Summary

Systematic survey from vehicle state estimation and trajectory tracking to collaborative CAV control applications.

Stop Avoidance

Frames comfort, energy saving, and transportation efficiency as central CAV control goals.

Collision Avoidance

Reviews vehicle-control approaches relevant to safety and collaborative control.

Validation

Literature survey rather than original experiments.

Limitations

Broad CAV control overview; not intersection-smoothing specific.

Use In This Project

Useful context citation for why trajectory tracking, collaborative control, safety, comfort, and energy belong together.

Venue/source note: Survey preprint; broad coverage, venue reputation not verified.

P14 · 2024 · Workshop · relevance 70/100

A Distributed Approach to Autonomous Intersection Management via Multi-Agent Reinforcement Learning

Matteo Cederle; Marco Fabris; Gian Antonio Susto

Venue
ATT 2024 Workshop, CEUR-WS Vol. 3813

Theme
Distributed MARL for AIM

Source
https://arxiv.org/abs/2405.08655

PDF
https://arxiv.org/pdf/2405.08655
open PDF downloaded from arXiv

Technical Summary

Distributed multi-agent RL for autonomous intersection management using 3D surround-view assumptions and prioritized scenario replay.

Stop Avoidance

Improves benchmark metrics in SMARTS, implying smoother flow through decentralized decisions.

Collision Avoidance

Aims for accurate decentralized navigation without a central server; safety is evaluated through simulation metrics.

Validation

SMARTS virtual environment; accepted at ATT 2024 and available through CEUR-WS.

Limitations

Workshop paper; not focused on explicit trajectory smoothing or formal safety constraints.

Use In This Project

Useful contrast to the proposed centralized JSSP/RL rollout, showing distributed AIM as an alternative branch.

Venue/source note: Peer-reviewed workshop proceedings in CEUR-WS; reputable but less archival than major journals.

P15 · 2023 · Preprint · relevance 73/100

D-HAL: Distributed Hierarchical Adversarial Learning for Multi-Agent Interaction in Autonomous Intersection Management

Guanzhou Li; Jianping Wu; Yujing He

Venue
arXiv preprint

Theme
Distributed hierarchical learning for AIM

Source
https://arxiv.org/abs/2303.02630

PDF
https://arxiv.org/pdf/2303.02630
open PDF downloaded from arXiv

Technical Summary

Non-RL adversarial actor/discriminator framework evaluates immediate interaction and final trajectory quality for multi-agent AIM.

Stop Avoidance

Targets reduced travel time and smoother interaction outcomes compared with distributed learning baselines.

Collision Avoidance

Immediate and final discriminators score interaction safety and trajectory outcomes.

Validation

Four-way six-lane intersection experiments against state-of-the-art methods.

Limitations

Learning framework lacks explicit optimization/smoothing formulation and formal safety guarantees.

Use In This Project

Useful for explaining why learned interaction policies must still be checked by a low-level feasibility/safety layer.

Venue/source note: arXiv preprint; venue reputation not verified.